

OCEAN ENTERPRISE

Unlocking Shared Value: The Ocean Enterprise Framework for Data and AI Ecosystems

Ocean Enterprise Collective e.V.

Whitepaper

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Abstract

Ocean Enterprise is an open-source software framework and governance model designed to create next-generation data spaces and AI ecosystems. It addresses challenges such as data silos, compliance, control, and value creation by enabling organizations to collaborate efficiently and create value from digital resources at scale without surrendering control. The framework utilizes a federated architecture that combines technical data sovereignty, value creation and exchange, transparency, sovereign deployments and Compute-to-Data capabilities that provide pier-to-pier access control to privately held data. This allows data to be used for computation and AI model training without the underlying intellectual property leaving its owner's control. Governed by the Ocean Enterprise Collective e.V., a non-profit association, it ensures transparent governance and alignment with key European data standards and regulations.

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1. Executive Summary

Data has become the most critical enabler of innovation and competitiveness in the digital economy, especially for AI. Yet most organizations still operate within isolated platforms, struggling to balance data availability, compliance, control, and value creation.

Ocean Enterprise changes this paradigm. It provides an open-source software framework for next generation dataspace and governance model that enables organizations to collaborate efficiently and create value from digital resources at scale — without surrendering control.

Ocean Enterprise offers a federated architecture that combines technical data sovereignty, value creation and exchange, transparency, sovereign deployments, and Compute-to-Data capabilities. This allows data to be used for computation, combined with software, to generate data products, train AI models and provide valuable insight without IP leaving its owner's control.

The framework is governed by **Ocean Enterprise Collective e.V.**, a registered non-profit association ("eingetragener Verein") in Germany. The association ensures transparent governance, legal integrity, and alignment with European standards for data spaces, digital markets and services.

Ocean Enterprise's unique strengths — decentralized trust, technical data sovereignty with Compute-to-Data, no single point of control or failure, and a real-time value transaction layer — position it as the next-generation data and AI ecosystem model for enterprises and public organizations seeking to create value through collaboration while preserving control.

2. The New Data Imperative

Over the past decade, data has been redefined from a passive asset to an active driver of competitive differentiation. The increasing use of artificial intelligence, digital twins and smart automation has created a huge demand for diverse, high-quality, and up-to-date data.¹ This shift has exposed the limitations of legacy data platforms, which were designed for an era of centralized control and internal analytics rather than the dynamic, collaborative and decentralized needs of the modern digital economy and an efficient single market.

While the evolution from data warehouses and lakes to cloud platforms offered scalability, it failed to address the fundamental challenges of inter-organizational collaboration. These centralized models reinforced vendor lock-in, monopolistic structures, and concentrated risk, while the core issues of fragmentation, data sovereignty and trust remained unresolved. In response, the market is undergoing a significant architectural paradigm shift towards decentralized and federated models. This evolution is evident in enterprise strategies such as

¹<https://www.emerald.com/intr/article/doi/10.1108/INTR-02-2024-0195/1251330/Artificial-intelligence-adoption-and-revenue>

the 'data mesh', which advocates decentralized data ownership within an organization and is now extending to relationships between organizations.²

This transition is being actively shaped by a new generation of policies and regulations. European initiatives such as Gaia-X signal a clear political and economic commitment to building a federated, sovereign, and interoperable data and AI infrastructure. Concurrently, major legislation acts that have recently come into force such as the EU Data Governance Act (June 2022), EU Data Act (Sept 2023) and AI Act (August 2024) are establishing a new legal framework. These frameworks are catalysts for innovation, creating market demand for architectures where sovereignty, interoperability, fair competition, shared value creation and transparency are embedded by design.³⁴

In this context, Ocean Enterprise is a critical technological enabler. It provides a production-ready blueprint and open-source software framework that bridges the gap between the vision of decentralized innovation and the practical requirements of enterprise-grade compliance. It offers organizations a clear pathway to participate in and build trusted data and AI ecosystems, enabling them to create value, share and utilize data insights without exposing underlying assets. This capability is unlocking new collaborative potential in domains such as manufacturing, finance, energy, healthcare, mobility, aerospace, automotive, science, and more.

The definition of competitive advantage has fundamentally changed. It no longer depends on merely possessing the most data, but on being an active and trusted participant in a secure data and AI ecosystem that enables efficient usage and value creation from data availability, besides new, data-driven business models.⁵ The next chapter will explore the significant challenges that have, until now, made participation in such an ecosystem a rather complicated and resource-intensive endeavor for most organizations.

²https://www.researchgate.net/publication/371700929_Tensions_in_Inter-Organizational_Data_Sharing_Findings_from_Literature_and_Practice

³ <https://digital-strategy.ec.europa.eu/en/factpages/data-act-explained>

⁴<https://www.europarl.europa.eu/topics/en/article/20230601STO93804/eu-ai-act-first-regulation-on-artificial-intelligence>

⁵ <https://www.tandfonline.com/doi/full/10.1080/00472778.2024.2379999>

3. The Problem



Figure 1: Foundational barriers of data collaboration: Data Silos; Regulatory Complexity; and Missing Tooling

3.1 The AI Imperative and the Data Underutilization Paradox

Contemporary enterprises operate within a significant paradox. While the strategic imperative to leverage artificial intelligence for competitive advantage has never been more pronounced, the requisite data remains largely inaccessible.⁶ Although data-driven organizations achieve productivity gains of over 10%, it is estimated that only 12% of available enterprise data is ever analyzed. This underutilization is most evident in inter-organizational collaboration, where combining diverse datasets has the greatest potential for transformative innovation and value creation.⁷

The effectiveness of advanced AI models is directly proportional to the volume and diversity of their training data. This creates a competitive imbalance, whereby large corporations can leverage vast proprietary datasets while SMEs are at a disadvantage due to their comparatively limited data pools. For these SMEs, sharing data through inter-organizational collaboration is

⁶<https://www.emerald.com/intr/article/doi/10.1108/INTR-02-2024-0195/1251330/Artificial-intelligence-adoption-and-revenue>

⁷ <https://publications.parliament.uk/pa/cm201516/cmselect/cmsctech/468/468.pdf>

essential for growth and competitiveness.⁸ Despite this clear incentive, a persistent state of data isolation persists. Valuable data assets remain isolated within organizational data silos, incurring significant economic costs estimated at \$3.1 trillion annually in lost productivity and unrealized revenue. These silos are the consequence of deeply rooted rational concerns that have historically lacked a viable solution.⁹

3.2 Foundational Barriers: Data sovereignty and regulatory complexity

The persistence of data silos is a defensive response to a series of foundational barriers that hinder inter-organizational data sharing. The primary impediment is the imperative to maintain data sovereignty and protect intellectual property (IP). An organization's data and business secrets represent a core competitive asset, and traditional data-sharing models necessitate the transfer of this asset, which is effectively a loss of control. Although legal agreements provide a contractual framework, they offer no technical guarantees against the unauthorized replication, misuse, or leakage of data after transfer. Most organizations are unwilling to accept the risk of exposing valuable IP.¹⁰

Compounding this issue is an increasingly complex and punitive regulatory landscape. Navigating this legal landscape is challenging, with substantial financial penalties for non-compliance. Legislative frameworks such as the EU's General Data Protection Regulation (GDPR), the Data Act and the AI Act impose stringent and often overlapping obligations on how data is handled, processed, and shared. The AI Act, for instance, stipulates rigorous standards for data quality and bias mitigation in AI training, while the Data Act introduces novel regulations governing data access.¹¹ The legal ambiguity and severe repercussions of non-compliance — with GDPR fines totaling \$1.8 billion in 2023 — create a chilling effect that strongly discourages collaboration.¹²

3.3 The technical vacuum and vendor lock-in effects

Barriers relating to IP risk and regulatory complexity are exacerbated by a critical technical vacuum in the enterprise ecosystem. There is a fundamental absence of technical tools designed to facilitate secure, compliant data and AI collaboration while preserving control and data sovereignty.¹³ Conventional solutions have proven inadequate. Centralized data sharing

⁸<https://scholarspace.manoa.hawaii.edu/server/api/core/bitstreams/61e6bcec-d348-4b2d-b127-3c387dfba7d2/content>

⁹ <https://www.dataversity.net/breaking-down-data-silos-for-digital-transformation-success/>

¹⁰ <https://pmc.ncbi.nlm.nih.gov/articles/PMC12375187/>

¹¹ <https://www.whitecase.com/insight-alert/data-act-eus-bid-ensure-fairness-digital-environment-and-competitive-data-market-has>

¹² <https://www.arivonix.ai/blog/what-is-data-monetization-and-why-it-matters/>

¹³ https://www.researchgate.net/publication/371700929_Tensions_in_Inter-Organizational_Data_Sharing_Findings_from_Literature_and_Practice

systems , for example, typically require data owners to upload their assets to a third-party platform. This model reintroduces counterparty risk and perpetuates vendor lock-in effects.¹⁴

Many enterprises also operate legacy systems and fragmented IT infrastructures that were not designed for secure, interoperable data exchange with external entities. This creates a threefold dilemma: powerful business incentives to share data for AI-driven innovation are constrained by the imperatives of IP protection and regulatory compliance, and there are no adequate technological instruments to bridge the gap. Without appropriate, sovereignty-preserving technology, inter-organizational data sharing becomes a high-risk, low-trust proposition that most organizations rationally avoid, leaving trillions of dollars in potential economic value unrealized.¹⁵

The central problem of the modern data economy is navigating these challenges. Resolving this issue requires a new approach that can technically enforce control and data sovereignty, prevent vendor lock-in and provide a clear path to regulatory compliance by design.

4. The Solution: The Ocean Enterprise Framework

Data has become the most critical enabler of innovation and competitiveness in the digital economy. Yet, most organizations still operate within isolated platforms, struggling to balance the imperatives of data sharing, regulatory compliance, and intellectual property protection. Ocean Enterprise is the strategic framework and open-source technology stack engineered to resolve these challenges. It enables organizations to collaborate securely, transparently, and at scale without surrendering control of their data or relying on centralized intermediaries.

A Federated Architecture for a New Data & AI Economy

The Ocean Enterprise framework introduces a federated architecture designed to dismantle the foundational barriers that create data silos. It provides a new model for inter-organizational collaboration by directly addressing the core enterprise concerns of data sovereignty, regulatory risk, and technological lock-in.

Traditionally businesses have opted to use centrally controlled **data platforms** — efficient but restrictive, and increasingly misaligned with the principles of sovereignty and resilience. Open **data and AI ecosystems**, in contrast, offer a new innovative approach that distributes control across participants who collaborate under shared governance and interoperability standards while keeping full control over IP and business secrets. Ocean Enterprise embodies this new model.

¹⁴ <https://www.jpmorgan.com/kinexys/content-hub/collective-intelligence-from-data-silos>

¹⁵ https://www.researchgate.net/publication/376392595_Data_Silos_-_A_Roadblock_for_AI_Ops

Preserving Data Sovereignty through Sovereign Deployments

The framework's primary principle is the preservation of control and sovereignty. It directly confronts the intellectual property risks associated with traditional data sharing by eliminating the need for data transfer (while this is still possible for use cases where transfer is the goal and/or required.). This is accomplished through its cornerstone capability, Compute-to-Data (C2D). This paradigm allows data to be used with applications that require secure computation and insight generation—such as the training of AI models—without the need that the underlying data ever leaves its owner's secure environment. In other words, algorithms are sent to the data and computation is performed locally or in trusted environments. With C2D only the resulting insights and data products are shared with the data consumer allowing for the data owner to maintain full control of their data. This model provides a technical guarantee of control, allowing organizations to monetize the value of their data assets without compromising the assets themselves.

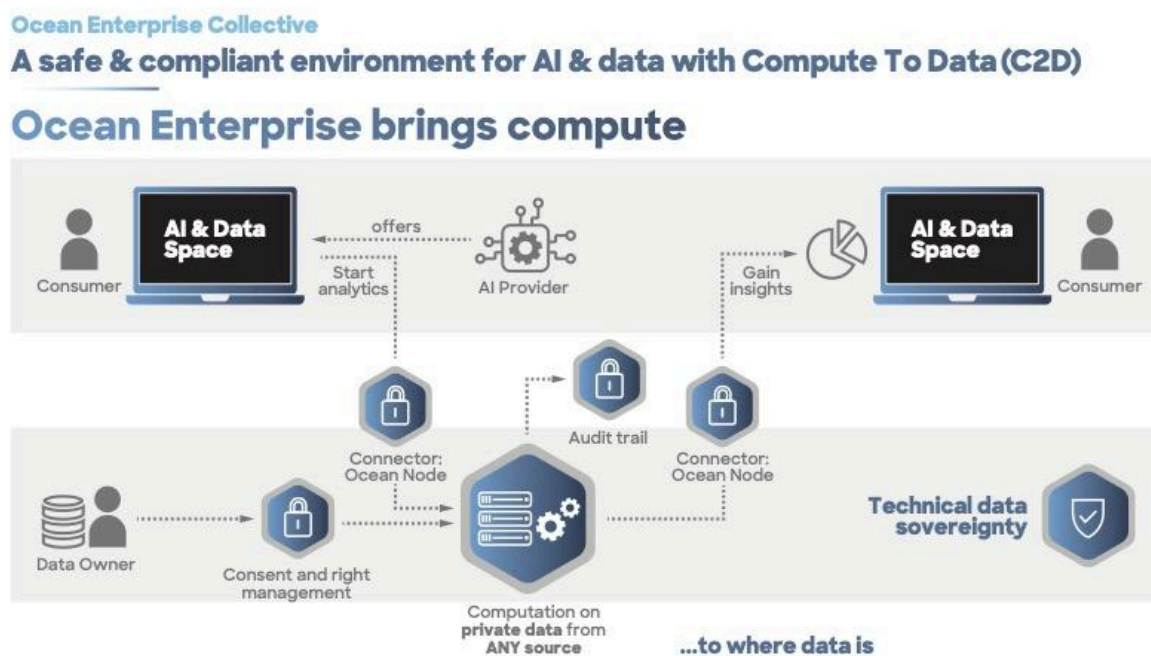


Figure 2: Compute-to-Data is a technology that allows computations to be performed directly on data without moving or exporting the raw data itself, thereby preserving privacy and data sovereignty. Only trusted applications are allowed to perform computation on the trusted infrastructure.

Ensuring Transparency and Compliance by Design

The Ocean Enterprise framework architecture is designed to navigate the complex regulatory landscape of the modern data and AI economy. Its decentralized infrastructure, based on smart contracts and Distributed Ledger Technology (DLT), creates a transparent and immutable audit

trail for all transactions. This provides verifiable proof of every interaction, which is essential for demonstrating compliance with regulations such as GDPR and the EU AI Act. Furthermore, the Compute-to-Data capability inherently supports data residency requirements by ensuring sensitive data remains within a specified jurisdiction. This "compliance-by-design" approach transforms regulatory adherence from a prohibitive barrier into a foundation of trust. Given full technical control, trust is inherently baked into the underlying architectural framework ..

Eliminating Lock-In Effects with Open, Decentralized Tooling

The Ocean Enterprise framework provides the missing tooling for a sovereign data and AI economy. As a free open-source software stack, it offers a compelling alternative to proprietary, centralized platforms, thereby mitigating the risk of vendor lock-in effects. Its federated architecture ensures there is no single point of control or failure, creating a resilient and neutral ecosystem for all participants. This open and interoperable nature lowers the economic and technical barriers to entry, enabling a more democratic and competitive data market.

A Strategic Framework for Value Creation

Beyond resolving the core problems of data sharing, the Ocean Enterprise framework also provides vital components necessary for building sustainable, value-generating data and AI ecosystems. It combines a robust governance model with a real-time value transaction layer to facilitate a new generation of data-driven business models and enhance existing business models through additional scalability.

Decentralized Trust and Governance

The framework is governed by the Ocean Enterprise Collective e.V., a registered non-profit association ("eingetragener Verein") in Germany. This governance structure ensures legal integrity, transparent stewardship of the open-source technology, and strategic alignment with European standards for data sharing and sovereignty. This model of decentralized governance, free from the influence of a single commercial entity, is critical for establishing the long-term trust required for inter-organizational collaboration.

A Real-Time Value Transaction Layer

The Ocean Enterprise framework enables a real-time value transaction layer for the data and AI economy. By automated contracting and the provisioning of access rights and licenses via smart contracts, it allows for the automated, peer-to-peer monetization of data, AI, algorithms, and other digital services. This creates a digital single and liquid market for data-driven services and aligns the incentives of all ecosystem participants, rewarding contributions and fostering a continuously growing and diversified network of services.

Aspect	Centralized Platform	1 st Generation Data Spaces	Ocean Enterprise Ecosystems
Ownership	Centralized, operated by a single provider	Centralized, mostly operated by a single provider	Federated, composed of sovereign participants
Governance	Proprietary and often opaque; controlled by the platform operator	Association or Consortium with a centralized Dataspace Authority	Transparent, managed by non-profit association
Data Control	Data leaves the owner's environment	Data can reside with the participants, but applications require data transfer.	Data remains within the owner's infrastructure via Compute-to-Data and data can be transferred if necessary.
Value Creation	Transactional; value captured primarily by the platform owner	Transactional, depending on the value streams allowed by the governance authority.	Collaborative, multi-party co-creation. Shared value creation through distributed participation and free competition.
Resilience	Vulnerable to single point of control	Vulnerable to a single point of control	Decentralized, no single point of control or failure
Trust Mechanism	Central authority. Based on platform reputation and terms of service. Strong lock-in.	Mostly central authority. Based on platform reputation and terms of service. Strong lock-in.	Based on cryptographic verification, self-sovereign identities, and open governance.
Vendor lock-in	High — switching providers entails high cost and re-integration effort	High — switching providers entails high cost and effort.	Limited — participants maintain independent deployments and interoperable interfaces
Competition	Possible as long as the platform provider does not interfere	Limited, depending on the platform provider and governance authority.	Unlimited to allow every participant to exploit their competitive advantages for the best products.
Information Asymmetries	The platform provider has internal insights and uses it to extract value.	The platform provider has internal insights and uses it to extract value.	Very limited, since information is shared between participants to enable a level playing-field.

The Ocean Enterprise framework and its unique strengths—decentralized trust, technical and verifiable data sovereignty through Compute-to-Data, and an open, federated architecture—position it as the next-generation data and AI ecosystem model. It provides the strategic and technical foundation for enterprises and organizations seeking to innovate through collaboration while preserving autonomy and control in the digital economy.

5. The Ocean Enterprise Framework Architecture

The Ocean Enterprise Framework is a multi-layered, open-source architecture engineered to resolve the foundational challenges of inter-organizational data collaboration. Its design provides a robust and flexible framework that enables technical data sovereignty, privacy-preserving computation, and decentralized monetization.

The **Ocean Enterprise Framework** provides a structured model for designing, governing, and participating in decentralized data ecosystems. It consists of five interconnected layers. Each layer performs a specific set of functions, working in concert to create a cohesive and secure data ecosystem.

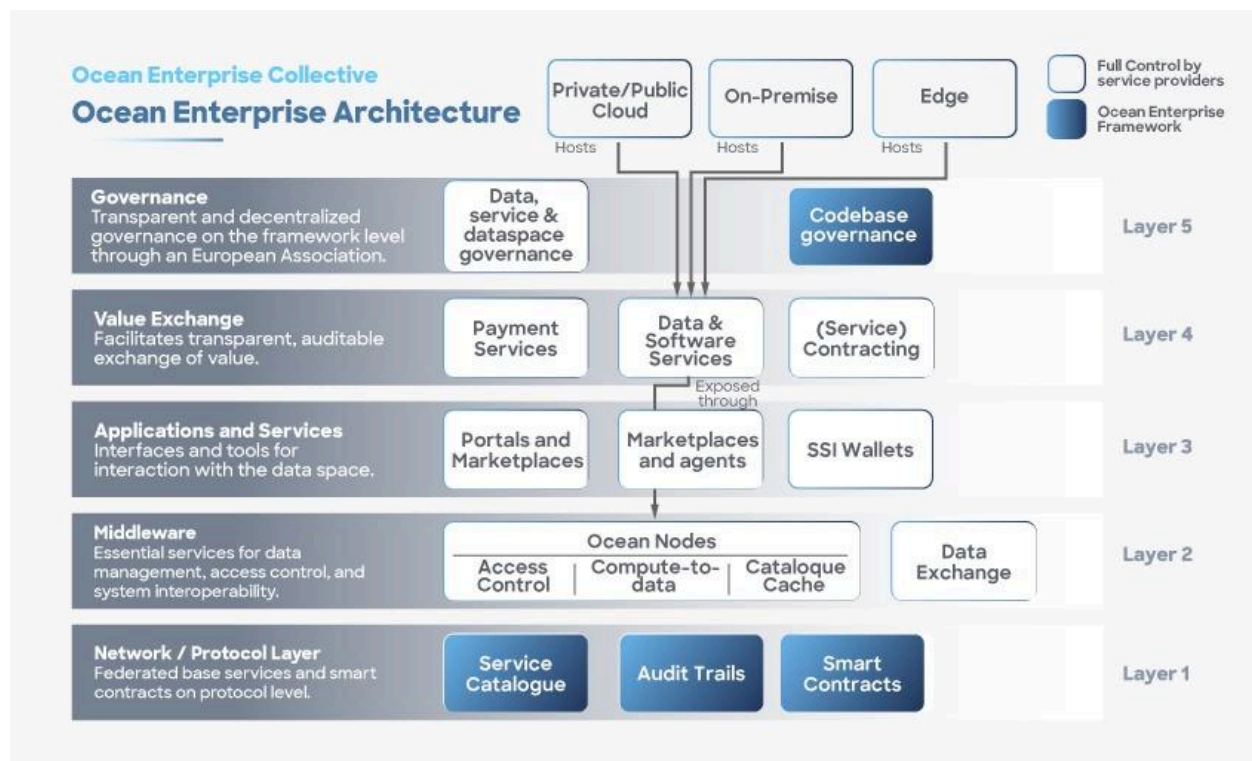


Figure 3: Ocean Enterprise Architecture

5.1 Layer 1: The Network and Protocol Layer

This foundational layer provides the decentralized infrastructure and core logic for the entire ecosystem. It establishes an immutable, trust-minimized environment where the rules of engagement are defined and enforced through cryptographic means rather than through reliance on a central intermediary. Its primary function is to serve as the "ground truth" for ownership, access rights, and all transactions within the data and AI economy.

Core Components and Functionality:

- **Decentralized Ledger Technology (DLT):** At its core, Ocean Enterprise architecture utilizes an Ethereum Virtual Machine (EVM)-compatible distributed ledger. This distributed ledger serves as a transparent and tamper-resistant distributed record that records all critical ecosystem interactions. Its decentralized nature eliminates single points of failure and control, ensuring high resilience and availability. For enterprise contexts that require increased confidentiality, this layer can be implemented on confidential EVMs or private EVMs, which encrypt smart contract states to protect sensitive business logic and transaction details and/or disclose information only to selected participants.
- **Smart Contracts (SC):** Deployed on the DLT, smart contracts are self-executing programs that automate and enforce the terms of an agreement between ecosystem participants. These contracts provide a comprehensive suite of functionalities that form the backbone of a sovereign data space:
 - **Federated Service Catalogue:** The Federated Service Catalogue is based on smart contracts and SC events, which store metadata of all resources available in the data space. These descriptions are anchored to Decentralized Identifiers (DIDs), a W3C standard, providing a tamper-resistant "single source of truth" for all service information. This functionality facilitates efficient resource discovery and utilization across the entire federated ecosystem without a central point of failure or control. Metadata is under the control of the service provider.
 - **Automatic Contracting and Sovereign Data Exchange:** The framework enables the decentralized creation, negotiation, and enforcement of legal exchange and usage agreements for any kind of service. It utilizes standardized, yet flexible contract templates, allowing participants to programmatically define and control the precise terms and conditions under which their data and AI services are used. This ensures technical sovereignty and is maintained at the technical and contractual level and integrates with middleware connectors to enforce access rights based on smart contract events.
 - **Authentication and Authorization:** The system implements robust data and service access through attribute-based access controls (ABAC). Access management relies on cryptographically signed messages and the exchange of Verifiable Credentials (VCs) linked to Self-Sovereign Identities (SSIs), ensuring high security and granular control over who can access specific resources, in line with the upcoming European Digital Identity (EUDI) standards.

- o Access Tokens: Smart contracts manage the creation of ERC-20 tokens that track contractually stipulated access rights to services (also known as “datatokens”). The use of this well-established standard enables detailed provenance tracking, creates immutable audit trails for all transactions, and ensures interoperability with a wide range of digital wallets and decentralized applications.
 - o Pricing and Settlement: Smart contracts facilitate fully automated and transparent monetization mechanisms. They enable the digitally verifiable collection and immediate distribution of fees to orchestrators, service providers, infrastructure operators, and other ecosystem participants. This system is highly customizable, allowing for the configuration of different fee collectors to support sustainable business models for all actors. It supports both ledger-based and off-ledger settlement options, including seamless integration with MiCA-compliant electronic money solutions like EURAU for real-time B2B payments.
- Tokenization of Intellectual Property (Data NFTs & Datatokens): The framework employs a sophisticated two-tiered token model to represent data services as programmable intellectual property.
 - o Data NFTs (ERC-721): Each unique data asset, algorithm, or digital service is first represented by metadata included in a Non-Fungible Token (NFT). This data or service NFT serves as the catalogue entry and base IP layer, establishing a singular, verifiable, and non-transferable record of ownership on the distributed ledger. Each NFT represents the base of a service offering to other dataspace ecosystem participants by a service provider. Since the metadata is owned and technically controlled by the service provider it cannot be altered, muted, or manipulated by anybody else.
 - o Access Tokens (ERC-20): The owner of the NFT has the exclusive right to mint fungible ERC-20 tokens, known as Access Tokens or datatokens. As detailed in the smart contract capabilities, each access token exchange represents the start of a specific license to access the underlying data or AI asset under contractually stipulated conditions. Each time an access token is exchanged a timestamped event is created that marks the beginning of a subscription to a service, based on a legal contract between two or more parties. This model transforms static data assets into liquid, programmable assets with granular, on-chain licensing capabilities. This model transforms static data assets into liquid, programmable assets with granular, on-chain licensing capabilities.

Layer 1 provides the fundamental guarantees of technical data sovereignty and transparency. By anchoring ownership and access rights to a decentralized and immutable ledger, it creates a secure foundation upon which all higher-level services can be built. Since Ocean Enterprise is DLT-agnostic there are no limitations where it can be used – in open or private ecosystems, depending on the industry.

5.2 Layer 2: The Middleware Layer

Acting as the bridge between the on-chain logic of the protocol layer and the user-facing applications, the middleware layer is the operational core of the stack. It translates the rules defined in the smart contracts into secure, real-world actions, managing data discovery, access control, and the execution of IP-preserving computation.

Core Components and Functionality:

- **Ocean Node:** Ocean Node is a unified software component that runs all core middleware services, replacing a suite of previously separate components. The Ocean Node is the primary engine that enforces the policies defined at the protocol layer and includes several critical functions:
 - **Indexer and Metadata Cache:** The node indexes metadata about data services from the distributed ledger and caches it in a high-performance database. This enables fast and efficient search and discovery of services within marketplaces, which is crucial for the usability of any data ecosystem.
 - **Policy Enforcement Point (Provider):** A core function of the Ocean Node is the Provider, a REST API that acts as the gatekeeper for data access. When a data consumer requests access, the Provider cryptographically verifies that a legal contract has been stipulated, and a subscription has been initiated before granting access to an AI or data service. This component is the technical enforcement point for data sovereignty, ensuring that no data or service is accessed without explicit, verified permission and acceptance of the terms and conditions.
- **Compute-to-Data (C2D) Orchestration:** The middleware layer manages the execution of the Compute-to-Data paradigm, the stack's cornerstone technology for privacy preservation. The C2D workflow is orchestrated by the Ocean Node's Provider component:
 - a) A data consumer sends an approved algorithm to the data provider's infrastructure.
 - b) The Provider initiates a secure compute environment, executes the algorithm on the local data, and monitors the process.
 - c) Upon completion, only the specified results (e.g., a trained AI model, statistical aggregates) are returned to the consumer. The raw data never leaves the data owner's secure control domain, thereby preserving privacy, protecting IP, and enabling compliance with data residency regulations like GDPR by design.
- **Developer Libraries:** To facilitate integration, the middleware layer includes a suite of powerful, open-source libraries such as `ocean.js` (TypeScript) and `ocean.py` (Python). These libraries provide well-documented APIs for developers to programmatically interact with the ecosystem, build custom applications, and integrate the protocol's functionalities into existing enterprise workflows. Implementations like Ocean CLI provide specialized participant agents for ecosystems that can be directly integrated with any other application to seamlessly connect to the data space ecosystem.

5.3 Layer 3: The Application and Services Layer

This is the layer where the protocol's capabilities are made tangible to end-users. It comprises vibrant and interoperable ecosystem data marketplaces, specialized service portals and identity wallets that facilitate user interaction and value creation.

Core Components and Functionality:

- **Data Marketplaces and Portals:** This layer hosts the user-facing interfaces, or participant agents, where data, AI and service providers can publish their offerings and consumers can discover, contract, and consume them. While public reference implementations exist, the primary enterprise application is the deployment of custom, domain-specific portals featuring networks of interconnected portals tailored for industries such as manufacturing, agriculture, smart cities, aerospace, and mobility. These markets can be custom-branded and curated while remaining fully interoperable with the broader federated ecosystem. They can be enriched with web-based applications and additional access controls, following the requirements of their industry.
- **"Everything-as-a-Service" (XaaS) Ecosystems:** The architecture is designed to support more than just data exchange. this layer can facilitate an "Everything-as-a-Service" model where participants can offer, discover, and monetize a wide array of digital services. This includes not only datasets but also software services, fully trained AI models, compute resources, and digital representations of physical assets like Asset Administration Shells (AAS) for Industry 4.0.
- **Decentralized Identity Wallets for Trust and Compliance:** This layer integrates decentralized identity wallets to create a complete, enterprise-ready solution that is secure, interoperable, and compliant with European regulations. This is achieved through a robust framework for identity, trust, and policy enforcement.
 - **Identity and Trust:** The ecosystem is built on asymmetric cryptography and Self-Sovereign Identity (SSI) wallets, which allows participants to manage their own digital identities without relying on any central authority. Specific attributes, rights, and qualifications are represented as Verifiable Credentials (VCs). The issuance of these VCs adheres to the OpenID for Verifiable Credential Issuance (OID4VCI) standard, an open protocol based on OAuth 2.0 for securely issuing digital credentials in a privacy-preserving manner. This entire identity framework is designed for compliance with the European eIDAS 2.0 regulation, ensuring that digital identities and attestations are legally recognized and interoperable across the EU.
 - **Policy Definition and Enforcement:** Usage policies and access rights are defined using standardized, machine-readable policy languages. This allows data owners to create granular and unambiguous usage contracts. The technical enforcement of these policies is handled by a dedicated policy engine. Implementations can utilize the Open Policy Agent (OPA) with its declarative language, Rego, or ODRL. These policy engines allow service providers and data spaces to extensively customize their usage rules and access restrictions.

- o **Holistic Integration:** These identity and policy components work in concert with other critical services. Gaia-X Digital Clearing Houses (GXDCH) can be used to automatically verify the compliance of participants and services against the well supported Gaia-X Trust Framework while data space ecosystems can integrate their own rules engines as required by their industry and target audience.

5.4 Layer 4: Value Exchange Layer

The Value Exchange Layer facilitates transparent, auditable exchange of value — whether monetary or non-monetary — through tokenized transactions, micropayments, or access-based incentives. Every data usage event can be tracked, priced, and settled in real time. For commercial transactions, the system integrates with regulated payment providers like EURAU to provide MiCA-compliant, and real-time B2B settlement for all services consumed within the ecosystem.

5.5 Layer 5: Governance Layer

Governed by **Ocean Enterprise Collective e.V.** on the framework level and ecosystem orchestrators on the ecosystem level, this layer defines policies, rights, and responsibilities. It ensures compliance with EU and global data regulations, enforces transparency, and prevents concentration of control.

6. Trust and Governance as the Differentiator

Ocean Enterprise is built on the principle that trust is engineered through transparent governance, rather than imposed through authority. Furthermore it reduces the trust that is needed to collaborate through technical data sovereignty and control.

The governance structure of **Ocean Enterprise Collective e.V.** ensures that no single organization controls the framework. Instead, it operates as a neutral, non-profit body that manages shared rules, and the evolution of the framework.

Ocean Enterprise Collective e.V.
VR41774, District court Berlin (Charlottenburg)

Key governance principles include:

- **Data Sovereignty:** Ownership and control remain with data originators.
- **Transparency:** All governance rules and changes are publicly available and verifiable.
- **Interoperability:** Open standards ensure cross-industry compatibility.
- **Accountability:** Roles, responsibilities, and compliance requirements are explicitly defined.
- **Resilience:** Decentralized infrastructure eliminates single points of failure.

This governance-first approach provides businesses with a **trusted operational foundation** — one that satisfies compliance obligations while enabling innovation across competitive and regulatory boundaries.

7. Value Realization and Strategic Differentiation

Ocean Enterprise transforms data collaboration into a **strategic differentiator** rather than a compliance exercise. Its design enables multiple forms of value creation:

Value Creation Models

1. **Collaborative Innovation:** Shared data access for AI model training, R&D, and product optimization.
2. **Operational Efficiency:** Compute-to-Data enables analytics on distributed datasets without central aggregation, reducing duplication and security risk.
3. **New Revenue Streams:** Real-time payments allow organizations to monetize data usage dynamically, with transparent pricing and settlement.
4. **Ecosystem Intelligence:** Cross-sector data collaboration enables insights that no single participant could achieve alone.
5. **Sustainable Data Economies:** Shared governance and transparent rules reduce vendor lock-in and build long-term ecosystem resilience.

These capabilities translate directly into **strategic agility** — enabling faster innovation, more resilient operations, and new business models built on data trust.

Data Regulation

In an era of intensifying data regulation and rising stakeholder expectations around trust and transparency, organizations must ensure that their data-sharing, monetization and collaborative data-ecosystem strategies are built on a foundation of rigorous compliance.

Ocean Enterprise transforms regulatory burdens into strategic assets by providing a secure, interoperable, and policy-aware data-sharing framework that aligns with the General Data Protection Regulation (GDPR) and the EU Data Act. Ocean Enterprise enables innovation rather than constraining it.

8. Pathways to Participation

Participation in Ocean Enterprise is structured to accommodate varying levels of maturity and ambition. Organizations can **start small** and scale their involvement as capabilities grow.

Recommended Executive Steps:

1. **Define the strategic intent** – Identify which business goals (e.g., innovation, efficiency, sustainability, Data Act compliance) ecosystem participation can advance.

2. **Assess readiness** – Review internal governance, data quality, and interoperability standards.
3. **Select role and partners** – Determine whether to act as provider, consumer, or orchestrator.
4. **Engage with Ocean Enterprise Collective e.V.** – Join as a member to shape governance and access the framework.
5. **Deploy sovereign nodes** – Operate Compute-to-Data capabilities within your own infrastructure.
6. **Launch pilot initiatives** – Validate value creation through a specific use case before scaling.

Participation does not require building a new ecosystem — it means joining a **trusted federation** where standards, governance, and technology are already aligned.

9. The Executive Agenda

Ocean Enterprise offers a path to create **new business models**, achieve **regulatory resilience**, and participate in cross-industry innovation networks.

It provides a **future-proof data ecosystem framework**— open, interoperable, and sovereign — capable of integrating with existing IT landscapes and compliance frameworks.

Together, leadership should:

- Position data sovereignty as a **strategic asset**, not a constraint;
- Embed **Compute-to-Data** principles into AI and analytics roadmaps;
- Invest in **interoperability and identity management** as shared capabilities;
- Participate actively in **Ocean Enterprise Collective e.V.** to influence standards and governance evolution.

The combination of **trust, decentralization, and real-time value exchange** establishes Ocean Enterprise as a **cornerstone of the emerging data and AI economy** towards an efficient, fair, and competitive digital single market for every organization and enterprise.

10. Conclusion

Ocean Enterprise represents the next evolution of data and AI ecosystems: trusted by design, sovereign by architecture, and built for real-time value creation.

Ocean Enterprise provides the blueprint for this future — combining the openness of decentralized technology with the rigor of enterprise-grade governance.

Through Ocean Enterprise, organizations can build on a trusted framework to exchange data, AI services, insights, and value — transparently, securely, and without dependency. The future of digital value creation is collaborative, not centralized.